

Statistics for Data Science

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These notes collect the statistics that data science courses assume.

Topics

The notes are organized in four groups of pages.

Describing data:

- [Nonparametric models](#): the empirical CDF, order statistics, and sample quantiles;
- [Exploratory data analysis](#): sample statistics, graphs, correlation, and contingency tables, illustrated with the WCGS data.

Inference:

- [Estimation](#): estimands, estimators, bias, mean squared error, and standard error;
- [Statistical inference](#): hypothesis tests, p-values, the t , chi-square, and F distributions, confidence intervals, and how to interpret negative findings;
- [Maximum likelihood inference](#): likelihood, score, information, the asymptotic distribution of MLEs, Wald and likelihood ratio tests, and Newton-Raphson, with worked examples.

Methods, illustrated with the HERS data:

- [Comparing means](#): t-tests and one-way ANOVA;
- [Comparing proportions](#): the chi-square test and Fisher's exact test;
- [Correlation and simple linear regression](#);
- [The bootstrap](#).

Bayesian:

- [Bayesian inference](#): the Bayesian paradigm, priors, posteriors, and hierarchical models;
- [Markov chain Monte Carlo](#): Monte Carlo integration, the Metropolis–Hastings and Gibbs samplers, convergence diagnostics, and the deviance information criterion;

- [Fitting models by Bayesian inference with JAGS](#): a proportion, logistic regression, survival, random effects, and Bayesian model averaging.

Origin

These notes began as the statistics appendices of *Regression Models for Epidemiology*, and keep those chapters' file names, except that two long chapters are now split into several pages. Most results keep their rme #id anchors on the same page; the exceptions are:

- the maximum likelihood practice exercises (`exr-prac-*`), which moved from the estimation page to the maximum likelihood page;
- the null and alternative hypothesis definitions, which moved from basic statistical methods to statistical inference;
- the sample statistics (mean, median, variance, standard deviation, interquartile range, proportion, correlation) and the contingency table, which rme defined on two pages; they are now defined once, on the exploratory data analysis page, under the ids rme used on its basic statistical methods page (such as `def-sample-mean`), so the exploratory-descriptive ids from rme `def-mean`, `def-median`, `def-eda-variance`, `def-eda-sd`, `def-quantile`, `def-iqr`, and `def-correlation` are retired;
- the basic statistical methods sections on categorical outcomes, correlation and simple linear regression, and the bootstrap (such as `sec-two-group-categorical`, `sec-correlation`, `sec-simple-linear-regression`, and `sec-bootstrap-ci`), which moved to their own pages, with their ids;
- the Bayesian inference sections on computation (`sec-foundations` through `sec-dic`) and the worked examples (`sec-bayes-examples` and the sections after it), which moved to the MCMC and JAGS pages, with their ids.

Where these notes rely on probability or calculus, they link to that book's [probability](#) and [mathematics](#) chapters.

Using these notes in another site

Course sites include these notes as a git submodule named `sds` at the site's root, and include fragments with paths that start with `sds/`, for example `{{< include sds/_subfiles/intro-MLEs/_def_mle.qmd >}}`. This site includes its own fragments the same way, through an `sds` symlink that points at the repository root.

Quarto resolves `@id` cross-references only within one rendered page, so a host site that links to a result here uses an explicit link, such as `[text](estimation.qmd#def-bias)`.